

## Skill lab

### Unit 3: musculoskeletal system

#### Week 3: Shoulder joint and radial nerve examination

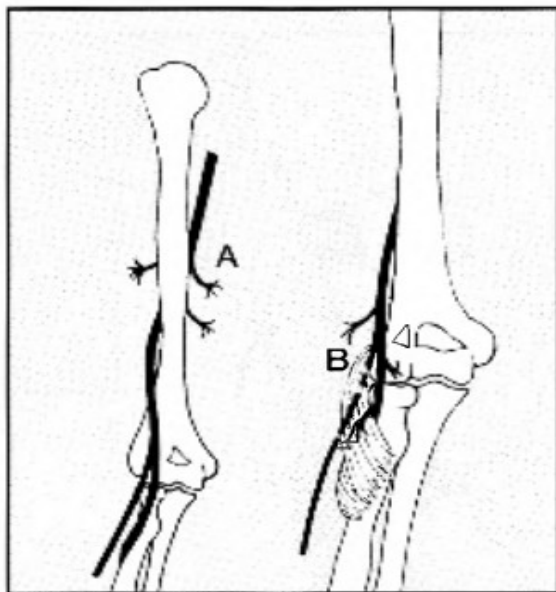
##### Objectives:

- 1- how to manage a patient presenting with trauma to the arm
- 2- how to perform physical examination of radial nerve
- 3- how to prescribe analgesics

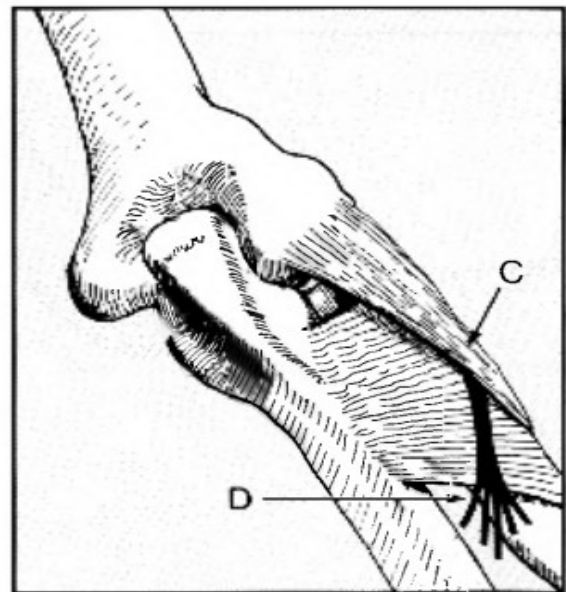
### Radial nerve

#### Examination sequence

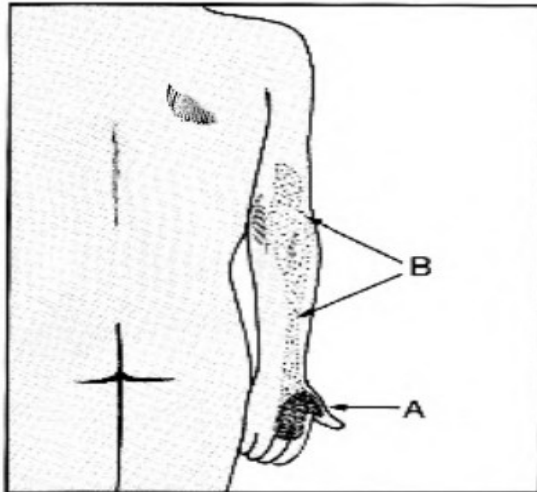
- Test for weakness of brachioradialis (elbow flexor) and the extensors of the arm (triceps). There will also be weakness of wrist and finger extension.
- Look for sensory loss over the dorsum of the hand and loss of triceps tendon jerk.



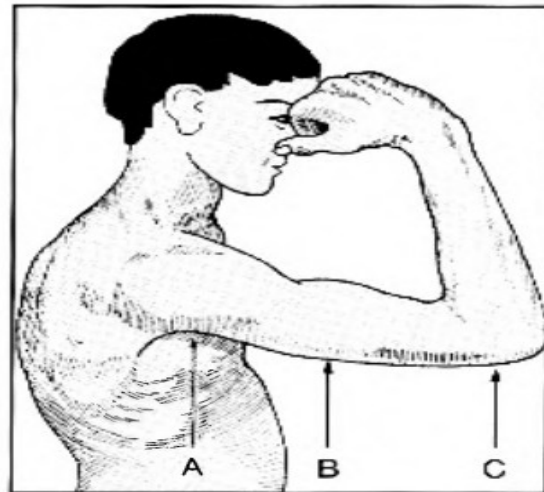
**2.29. Radial nerve: (posterior cord) C5,6,7,8 (T1):** *Motor distribution:* (A) In the upper arm the radial nerve supplies triceps. (B) In front of the elbow, it supplies brachioradialis, extensor carpi radialis longus and brachialis. Its posterior interosseous branch, before it enters the supinator tunnel, supplies extensor carpi radialis brevis and part of supinator.



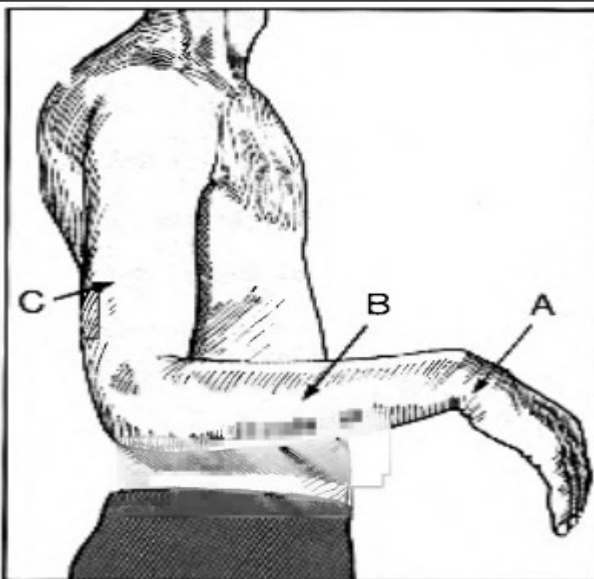
**2.30. Radial nerve (motor distribution) ctd.:** (C) In the supinator tunnel the posterior interosseous branch of the radial supplies the rest of supinator; lesions here may cause local tenderness. (D) On leaving supinator below the elbow it supplies extensor digitorum communis, extensors digiti minimi and indicis, extensor carpi ulnaris, abductor pollicis longus, and extensors pollicis longus and brevis.



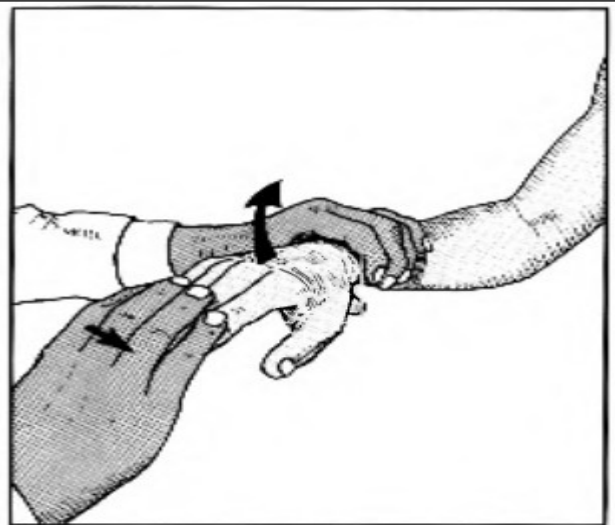
**2.31. Radial nerve: Sensory distribution:** (A) The terminal part (the superficial radial nerve) supplies the radial side of the back of the hand. (B) The posterior cutaneous branch of the radial, given off in the upper part of the arm, supplies a variable area on the back of the arm and forearm.



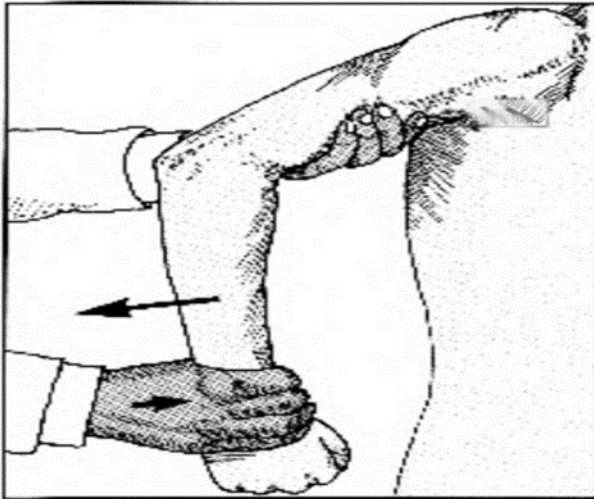
**2.32. Radial nerve: Common sites affected:** (A) In the axilla (e.g. from crutches, or the back of a chair in the so-called 'Saturday night' palsy); (B) Mid-humerus (from fractures and tourniquet palsies); (C) at and below the elbow (e.g. after dislocations of the elbow, Monteggia fractures, ganglions, and sometimes surgical trauma following exposures in this region).



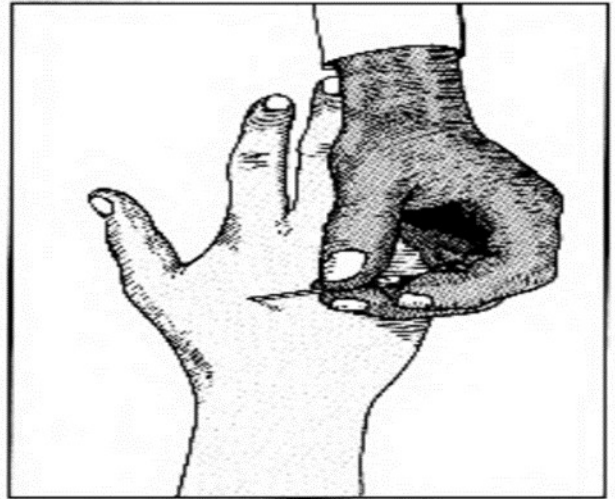
**2.33. Examination of the radial nerve (1):** Note in particular the following: (A) Is there an obvious wrist drop? (B) Is there wasting of the forearm muscles? (C) Is there wasting of the triceps, suggesting a high (proximal) lesion?



**2.34. Examination of the radial nerve (2):** Test the extensors of the wrist and fingers. The elbow should be flexed and the hand placed in pronation. Support the wrist, and ask the patient first to try and straighten the fingers and then to pull back the wrist; if there is any activity, judge the strength by applying counterpressure on the fingers or hand.



**2.37. Examination of the radial nerve (5):** Now test the triceps. Extend the shoulder and ask the patient to extend the elbow, first against gravity and then against resistance. Weakness of triceps suggests a lesion at mid-humeral level, or an incomplete high lesion; loss of all triceps activity suggests a high (plexus) lesion.



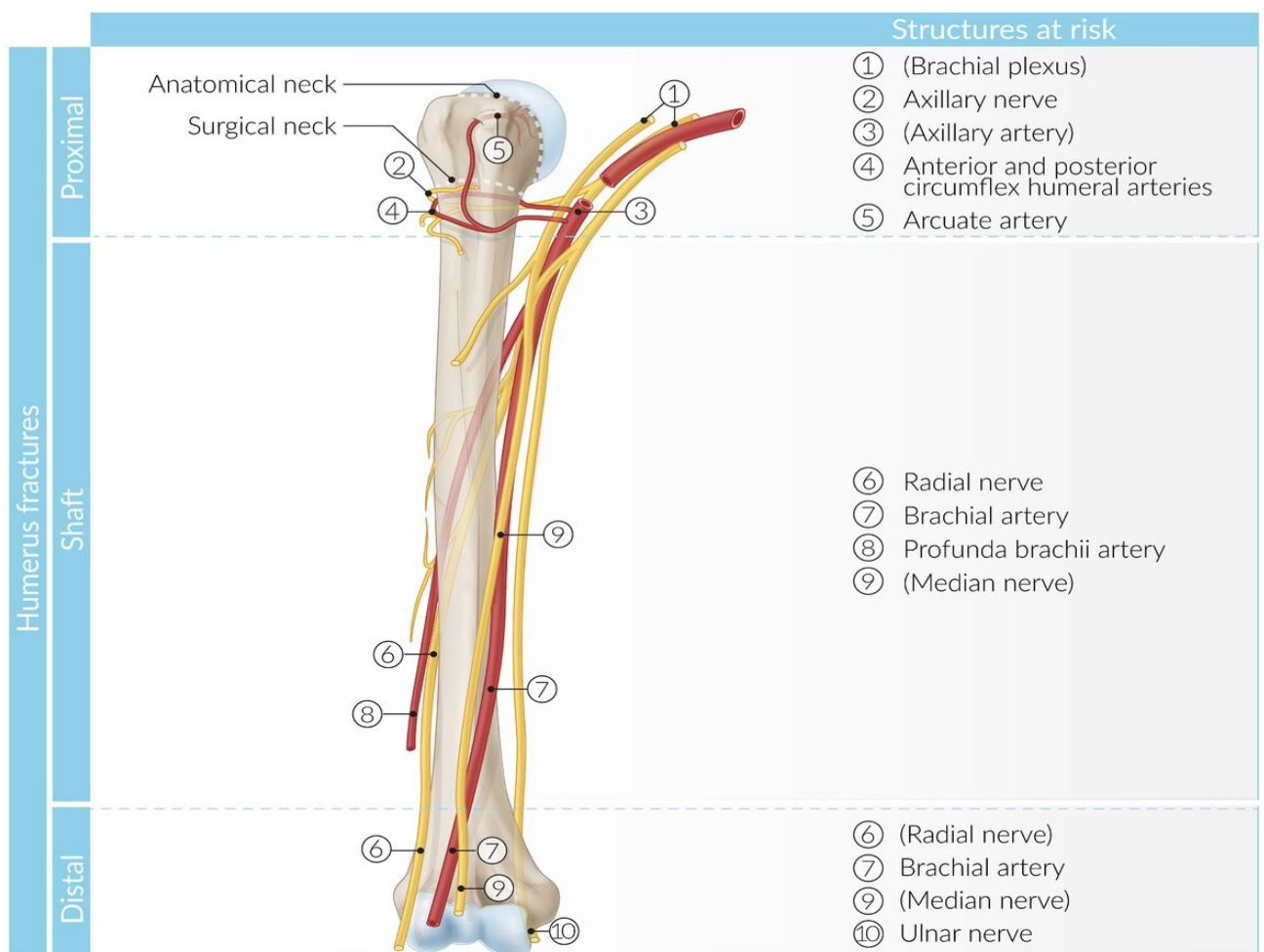
**2.38. Examination of the radial nerve (6):** Test for sensory loss in the areas supplied by the nerve. Loss confined to the hand indicates that the lesion is unlikely to be much proximal to the elbow. Careful analysis of both the motor and sensory deficits should allow accurate localization of the lesion.

**Case scenario: you are the resident in the emergency room to evaluate 67-year-old female with left arm pain for 3 hrs. duration. Manage the clinical situation dictates**

**Dear tutors: make sure that students understand that management does not start by dealing directly with fracture (ABCDE), passive movement here is CI, and should focus on neurovascular bundle exam particularly radial nerve and deep brachial artery**

**SP instruction**

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario</b>                              | Nidhal Hassan, 67 year old female                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>History of present illness</b>            | Chief complaint: left upper arm pain                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Duration                                     | 3 hrs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Precipitating events and mechanism of injury | Falling on outstretched hand while walking in the garden                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Aggravating/alleviating factors              | Constant pain aggravated by motion                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Character                                    | sharp                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Severity                                     | Sever 9/10                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Course                                       | progressive                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Associated symptoms                          | Swelling and bluish discoloration of right upper arm. No loss of consciousness. No other symptoms.                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Current life situation</b>                | (where does patient live/work, ...): retired teacher                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Personality</b>                           | In distress                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Past medical history</b>                  | hypertension for 10 years. Ischemic heart disease for 3 years                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Family history</b>                        | Her sister had multiple myeloma                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Social history</b>                        | No smoking. Not alcoholic                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Medications</b>                           | Oral antihypertensive, antiplatelet, lipid lowering agents. No drug allergy                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Physical exam</b>                         | Joint examination reveals moderate swelling, and extensive ecchymosis of the upper arm which is held in flexion. Tenderness at the midpart of right upper arm. Pain and limitation of active motion of the shoulder joint. Passive motion??<br>Normal cervical and elbow joint examination.<br>Normal brachial and radial pulses.<br>Radial nerve palsy: inability to extend the wrist and MCP joints with sensory loss over the snuffbox. Normal median and ulnar nerves. |



## Pathological anatomy

What is the difference between the following 2 pictures?



With fractures above the deltoid insertion, the proximal fragment is adducted by pectoralis major. With fractures lower down, the proximal fragment is abducted by the deltoid. Injury to the radial nerve is common, though fortunately recovery is usual.

**65 year old female with hypertension and ischemic heart disease presents with right arm pain and swelling after falling. She was diagnosed as having humerus shaft fracture. Prescribe an analgesic for her.**